Cryptanalysis of Hecke-KE: A Linear-Algebra Attack via Hecke Eigenbasis Decomposition

Xiyao Chen

University of Wisconsin–Madison
xiyaochen2002@gmail.com

Abstract. We give a complete passive attack on the Hecke-KE key-exchange scheme. The scheme proposes using products of Hecke operators on $S_k(\Gamma_0(N))$ as a one-way function. We show that the Hecke algebra acting on any fixed $S_k(\Gamma_0(N))$ is simultaneously diagonalizable over an explicit number field computable from the public parameters alone, and that this diagonalization reduces shared-key recovery to d scalar divisions over that number field, where $d = \dim S_k(\Gamma_0(N))$. For the benchmark parameter sets of [1] we have $d \leq 7$; our main theorem shows that enlarging d does not rescue the scheme. The precomputation is a one-time public computation (eigenbasis of $S_k(\Gamma_0(N))$, costing $\widetilde{O}(B \cdot d^3)$ rational operations, where $B = O(N)$ is the Sturm bound); the per-session attack cost is then $O(d^2)$ field operations, entirely independent of the pool size r and the number of Hecke factors s. We verify the attack in SageMath 10.7 against all parameter sets from the paper; in every case the recovered key satisfies $K' = K$. Furthermore, we prove that the attack runs in time polynomial in $d = \dim S_k(\Gamma_0(N))$ for *every* level N (prime or composite) and every weight k, while the honest protocol's public-key size is $\Omega(d)$ rationals. Consequently there is no choice of (N, k) for which Hecke-KE is secure and implementable: the scheme is unfixable within its design framework.

Keywords: Hecke operators · modular forms · key exchange · cryptanalysis · post-quantum cryptography

1 Scheme Recap

We recall the Hecke-KE protocol of [1] using the same notation.

Public parameters. Fix a prime N, a weight $k = 2$, and let $M = S_k(\Gamma_0(N))$ be the space of weight-k cusp forms for $\Gamma_0(N)$, of dimension d. Choose a public "distinguishing" form $f \in M$ (a generic $\mathbb{Q}$-linear combination of the standard basis). Let $S_A = \{p_1, \ldots, p_r\}$ and $S_B = \{q_1, \ldots, q_r\}$ be two disjoint sets of primes, all coprime to N.

Key generation. Alice samples s squarefree integers $N_1, \ldots, N_s$ (each a product of three distinct primes from S_A), sets $H_A = T_{N_1} \circ \cdots \circ T_{N_s}$, and publishes $H_A(f)$. Bob samples analogous integers $M_1, \ldots, M_s$ from S_B, sets $H_B = T_{M_1} \circ \cdots \circ T_{M_s}$, and publishes $H_B(f)$.

Shared secret. Both parties apply their own operator to the other's public message:

$$K_A = H_A(H_B(f)), \qquad K_B = H_B(H_A(f)).$$

Since Hecke operators commute (Theorem 1), $K_A = K_B =: K$.

Security claim. Li [1] bases security on the hardness of the *Hecke computational Diffie–Hellman* (Hecke-CDH) problem: given $(f, H_A(f), H_B(f))$, compute $H_A(H_B(f))$. The paper further argues that a "natural attack" using the eigenbasis is fundamentally limited because it leads to an underdetermined system. We show this argument is incorrect.

Coefficient domain. The public messages $H_A(f)$ and $H_B(f)$ are coordinate vectors over $\mathbb{Q}$ in the standard basis of $S_k(\Gamma_0(N))$; in general these coordinates are rational numbers, not integers. Our experiments confirm this: for $N = 53$ the shared secret contains coordinates such as $-1133949902748800935202816 0/3$. Hecke operators preserve the $\mathbb{Q}$-structure of $S_k(\Gamma_0(N))$ but not the $\mathbb{Z}$-lattice defined by any particular basis. Consequently, the public-key size bound of [1, Table 1] — stated in bytes and implicitly assuming integer coordinates — is a strict underestimate: the actual bit-length of a rational coordinate written as a reduced fraction p/q may substantially exceed the bound quoted there.

2 Mathematical Background

This section collects the definitions and classical results from the theory of modular forms that are used in the attack. We do not assume prior familiarity with the subject. Standard references are Diamond–Shurman [3] and Stein [4].

2.1 Modular Forms and Cusp Forms

Let $\mathbb{H} = \{\tau \in \mathbb{C} : \mathrm{Im}(\tau) > 0\}$ be the complex upper half-plane. The group $\mathrm{SL}_2(\mathbb{Z})$ acts on $\mathbb{H}$ by Möbius transformations $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \cdot \tau = \frac{a\tau+b}{c\tau+d}$.

Definition 1 (Congruence subgroup). *For a positive integer N (the* level*),*

$$\Gamma_0(N) \;=\; \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}) : c \equiv 0 \pmod{N} \right\}.$$

Definition 2 (Modular form and cusp form). *Let k be a positive even integer (the* weight*). A holomorphic function $f : \mathbb{H} \to \mathbb{C}$ is a* modular form of weight k for $\Gamma_0(N)$ *if:*

1. *$f\left(\frac{a\tau+b}{c\tau+d}\right) = (c\tau+d)^k f(\tau)$ for all $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \Gamma_0(N)$;*
2. *f is holomorphic at every* cusp *of $\Gamma_0(N)$ (the $\mathbb{Q} \cup \{i\infty\}$-orbits under the $\Gamma_0(N)$-action).*

A modular form is a cusp form *if it vanishes at every cusp. The $\mathbb{C}$-vector space of cusp forms of weight k for $\Gamma_0(N)$ is denoted $S_k(\Gamma_0(N))$.*

Every cusp form has a *Fourier (q-expansion)* at the cusp $i\infty$:

$$f(\tau) = \sum_{n=1}^{\infty} a_n(f)\, q^n, \qquad q = e^{2\pi i \tau},$$

with $a_n(f) \in \mathbb{C}$. The space $S_k(\Gamma_0(N))$ has an explicit $\mathbb{Q}$-rational basis (the *Victor Miller basis*) whose q-expansions are computable to any desired precision.

Proposition 1 (Dimension of $S_k(\Gamma_0(N))$ [4, Ch. 6]). *For prime level $N \geq 5$ and weight $k = 2$,*

$$d \;:=\; \dim S_2(\Gamma_0(N)) \;=\; g(X_0(N)),$$

the genus of the modular curve $X_0(N)$. For any prime p and any positive integer N, let $v_p(N)$ be the power of p that divides N. Also, let

$$\mu_0(N) = \prod_{p \mid N} \left(p^{v_p(N)} + p^{v_p(N)-1} \right), \quad \mu_{0,2}(N) = \begin{cases} 0 & \text{if } 4 \mid N, \\ \prod_{p \mid N} \left(1 + \left(\frac{-4}{p}\right)\right) & \text{otherwise,} \end{cases}$$

$$\mu_{0,3}(N) = \begin{cases} 0 & \text{if } 2 \mid N \text{ or } 9 \mid N, \\ \prod_{p \mid N} \left(1 + \left(\frac{-3}{p}\right)\right) & \text{otherwise,} \end{cases}$$

$$c_0(N) = \sum_{d \mid N} \varphi(\gcd(d, N/d)),$$

$$g_0(N) = 1 + \frac{\mu_0(N)}{12} - \frac{\mu_{0,2}(N)}{4} - \frac{\mu_{0,3}(N)}{3} - \frac{c_0(N)}{2}.$$

Note that $\mu_0(N)$ is the index of $\Gamma_0(N)$ in $\mathrm{SL}_2(\mathbb{Z})$.
Then we have $\dim S_2(\Gamma_0(N)) = g_0(N)$, and for $k \geq 4$ even,

$$\dim S_k(\Gamma_0(N)) = (k-1) \cdot (g_0(N) - 1) + \left(\frac{k}{2} - 1\right) \cdot c_0(N) + \mu_{0,2}(N) \cdot \left\lfloor \frac{k}{4} \right\rfloor + \mu_{0,3}(N) \cdot \left\lfloor \frac{k}{3} \right\rfloor.$$

The dimension of the Eisenstein subspace is

$$\dim E_k(\Gamma_0(N)) = \begin{cases} c_0(N) & \text{if } k \neq 2, \\ c_0(N) - 1 & \text{if } k = 2. \end{cases}$$

2.2 Hecke Operators

Definition 3 (Hecke operator). *For a positive integer n with $\gcd(n, N) = 1$, the* Hecke operator $T_n : S_k(\Gamma_0(N)) \to S_k(\Gamma_0(N))$ *is defined by its action on Fourier coefficients:*

$$a_m(T_n f) \;=\; \sum_{d \mid \gcd(m,n)} d^{k-1}\, a_{mn/d^2}(f).$$

For a prime $p \nmid N$ this specialises to

$$a_m(T_p f) \;=\; a_{mp}(f) + p^{k-1} a_{m/p}(f),$$

where $a_{m/p}(f) = 0$ when $p \nmid m$. For prime powers and composites, T_n is defined by the multiplicative extension $T_{mn} = T_m T_n$ ($\gcd(m, n) = 1$) and the recurrence $T_{p^r} = T_p T_{p^{r-1}} - p^{k-1} T_{p^{r-2}}$.

Concretely, once a basis $\{b_1, \ldots, b_d\}$ of $S_k(\Gamma_0(N))$ is fixed, each T_n is represented by a $d \times d$ integer matrix (the *Hecke matrix*), computable via the q-expansion action above. The Hecke matrices for all primes $p \leq B$ (Sturm bound, Definition 5) suffice to determine T_n for all n by multiplicativity.

Theorem 1 (Properties of Hecke operators [3, Ch. 5]). *The operators $\{T_n : \gcd(n, N) = 1\}$ on $S_k(\Gamma_0(N))$ satisfy:*

1. Commutativity*: $T_m T_n = T_n T_m$ for all m, n with $\gcd(mn, N) = 1$.*
2. Multiplicativity*: $T_{mn} = T_m T_n$ whenever $\gcd(m, n) = 1$.*
3. Self-adjointness*: Each T_n is self-adjoint with respect to the Petersson inner product*

$$\langle f, g \rangle = \int_{\Gamma_0(N)\backslash\mathbb{H}} f(\tau)\overline{g(\tau)}\, y^{k-2}\, dx\, dy$$

.

The $\mathbb{Z}$-algebra $\mathbb{T} = \mathbb{Z}[T_n : \gcd(n, N) = 1]$ is called the *Hecke algebra* of $S_k(\Gamma_0(N))$. Theorem 1(3) implies $\mathbb{T}$ is a commutative semisimple $\mathbb{Q}$-algebra (every self-adjoint operator on an inner product space over $\mathbb{C}$ is diagonalizable with real eigenvalues, and commuting diagonalizable operators are simultaneously diagonalizable).

2.3 Newforms and Galois Orbits

Definition 4 (Newform). *A cusp form $f \in S_k(\Gamma_0(N))$ is a* newform *if:*

1. *f is a normalised ($a_1(f) = 1$) simultaneous eigenform for all T_n with $\gcd(n, N) = 1$;*
2. *f is orthogonal (under the Petersson inner product) to all* oldforms*, i.e. forms of the type $g(\ell\tau)$ with $g \in S_k(\Gamma_0(M))^{\text{new}}$ and $M\ell \mid N$, $M < N$.*

If $T_n f = \lambda_n f$, the scalar $\lambda_n \in \mathbb{C}$ is the n-th Hecke eigenvalue *of f. The* eigenvalue field *of f is $K_f = \mathbb{Q}(\lambda_p : p \nmid N) \subset \mathbb{C}$; this is always a number field (finite extension of $\mathbb{Q}$) [3, Thm. 5.8.3].*

The absolute Galois group $\mathrm{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$ acts on the set of newforms in $S_k(\Gamma_0(N))$ by acting on their q-expansion coefficients. The orbits of this action are called *Galois orbits* of newforms.

Theorem 2 (Atkin–Lehner newform decomposition [2]). *For prime level N, the space $S_k(\Gamma_0(N))$ has an orthogonal decomposition*

$$S_k(\Gamma_0(N)) \;=\; \bigoplus_{[f]} V_{[f]},$$

indexed by the Galois orbits $[f]$ of newforms. Each Galois-orbit subspace $V_{[f]}$ satisfies $\dim V_{[f]} = [K_f : \mathbb{Q}] =: e_{[f]}$, and $\sum_{[f]} e_{[f]} = d$. Over $\bar{\mathbb{Q}}$, each orbit decomposes further into one-dimensional eigenspaces:

$$V_{[f]} \otimes_{\mathbb{Q}} \bar{\mathbb{Q}} \;=\; \bigoplus_{\sigma \in \mathrm{Hom}(K_f, \bar{\mathbb{Q}})} \bar{\mathbb{Q}} \cdot \sigma(f),$$

where $\sigma(f)$ denotes the newform whose q-expansion has coefficients $\sigma(\lambda_n)$.

Corollary 1 (Simultaneous diagonalizability). *Let*

$$K \;=\; \mathrm{comp}\big(\tilde{K}_f : [f] \text{ a Galois orbit of newforms in } S_k(\Gamma_0(N))\big),$$

where $\tilde{K}_f$ denotes the Galois closure of K_f over $\mathbb{Q}$ (so $\tilde{K}_f = K_f$ whenever $K_f/\mathbb{Q}$ is already Galois). Then $S_k(\Gamma_0(N)) \otimes_{\mathbb{Q}} K$ has a basis $\{v_1, \ldots, v_d\}$ of common eigenvectors for all Hecke operators:

$$T_n(v_j) \;=\; \lambda_{n,j}\, v_j \qquad \forall\, n \text{ with } \gcd(n, N) = 1, \quad j = 1, \ldots, d,$$

with $\lambda_{n,j} \in K$. The eigenvalues $\{\lambda_{n,j}\}_{j=1}^{d}$ are pairwise distinct for some prime $n \nmid N$ (by the Multiplicity One property of newforms [3, Thm. 5.8.2(b)]), so the eigenvectors $v_1, \ldots, v_d$ are K-linearly independent. Consequently, in the standard basis the $d \times d$ change-of-basis matrix V (columns $= v_j$) is invertible and diagonalises every Hecke matrix:

$$T_n \;=\; V \cdot \mathrm{diag}(\lambda_{n,j}) \cdot V^{-1}.$$

The matrix V is computable from (N, k) alone (see Sect. 4).

2.4 Sturm Bound

Definition 5 (Sturm bound [4, Thm. 9.18]). *The* Sturm bound *for $S_k(\Gamma_0(N))$ is*

$$B \;=\; \left\lfloor \frac{k}{12}\,[\mathrm{SL}_2(\mathbb{Z}) : \Gamma_0(N)] \right\rfloor + 1 \;=\; \left\lfloor \frac{k(N+1)}{12} \right\rfloor + 1 \quad \textit{(for prime } N\textit{)}.$$

Two cusp forms in $S_k(\Gamma_0(N))$ are equal if and only if their q-expansions agree through degree B. In particular, expressing any modular form in the standard basis of $S_k(\Gamma_0(N))$ reduces to solving a $B \times d$ linear system. For the paper's parameters ($k = 2$, $N \le 97$) we have $B \le 17$.

3 Key Observation: Simultaneous Diagonalizability

Corollary 1 is the conceptual heart of the attack. We restate it in the protocol's notation for clarity.

The Hecke algebra $\mathbb{T}$ acting on $S_k(\Gamma_0(N))$ is a finite-dimensional, commutative, semisimple $\mathbb{Q}$-algebra (semisimplicity follows from self-adjointness, Theorem 1(3)). By Corollary 1 there exists an explicit number field K and an invertible $d \times d$ matrix V over K such that

$$T_n \;=\; V \cdot \mathrm{diag}(\lambda_{n,1}, \ldots, \lambda_{n,d}) \cdot V^{-1} \qquad \forall\, n \text{ with } \gcd(n, N) = 1. \tag{1}$$

For any composite Hecke operator $H = T_{N_1} \circ \cdots \circ T_{N_s}$,

$$H \;=\; V \cdot \mathrm{diag}(\Lambda_1, \ldots, \Lambda_d) \cdot V^{-1}, \qquad \Lambda_j = \prod_{i=1}^{s} \lambda_{N_i,j}. \tag{2}$$

The scalar Λ_j is the eigenvalue of H on the j-th eigenline. Crucially, V depends only on (N, k) and is *publicly computable*.

4 The Attack

The attack uses only the public information $(N, k, f, H_A(f), H_B(f))$ and proceeds in four steps. Pseudocode appears as Algorithms 1 and 2.

Step 1: Compute the eigenbasis. Let $\{b_1, \ldots, b_d\}$ be the standard basis of $S_k(\Gamma_0(N))$. For each Galois orbit $[f_\ell]$ of degree e_ℓ with eigenvalue field K_ℓ:

1. Compute the q-expansion of f_ℓ over K_ℓ up to precision $\geq B$ (Definition 5); in our implementation we use precision $B + 5$ as a safety margin.
2. Build the $B \times d$ matrix A_ℓ over K_ℓ with $(A_\ell)_{ij} = [q^i](b_j)$. Solve the consistent overdetermined system $A_\ell c = y_\ell$ ($y_\ell = (a_1(f_\ell), \ldots, a_B(f_\ell))$) over K_ℓ to obtain $c \in K_\ell^d$, the standard-basis coordinate vector of f_ℓ.
3. Obtain all e_ℓ Galois-conjugate eigenvectors. If $K_\ell/\mathbb{Q}$ is Galois, apply each automorphism $\phi \in \mathrm{Aut}(K_\ell/\mathbb{Q})$ coordinate-wise to c. Otherwise let L_ℓ be the Galois closure of K_ℓ and use the e_ℓ distinct embeddings $\phi : K_\ell \hookrightarrow L_\ell$ to form $\phi(c) \in L_\ell^d$.

This yields $\sum_\ell e_\ell = d$ eigenvectors, one per eigenline. By Corollary 1 these are K-linearly independent; arrange them as the d columns of a $d \times d$ matrix V over $K = \mathrm{comp}(L_1, \ldots, L_r)$, and compute V^{-1}.

Remark 1 (Oldform fallback). Algorithm 1 as stated uses the newform q-expansion approach, which applies whenever $\sum_{[f_\ell]}[K_{f_\ell} : \mathbb{Q}] = d$ (i.e. the newforms at level N already span $S_k(\Gamma_0(N))$). This holds for all prime levels considered in [1]. For composite levels, $S_k(\Gamma_0(N))$ may contain oldforms that are not spanned by level-N newforms. In that case one replaces Step 1 with a direct Hecke-matrix eigenvector computation: decompose $S_k(\Gamma_0(N))$ into Hecke-simple submodules (available in SageMath as `M.decomposition()`), then for each block find a prime $p \nmid N$ such that the restriction of T_p has a squarefree characteristic polynomial, adjoin the root as a number field, and compute the eigenvector by kernel computation over that field. All subsequent steps are identical; the attack's correctness and complexity bounds are unchanged.

Step 2: Project into the eigenbasis. Let $v_f, v_A, v_B \in \mathbb{Q}^d$ be the coordinate vectors of f, $H_A(f)$, $H_B(f)$ in the standard basis. Compute

$$\hat{f} = V^{-1} v_f, \qquad \hat{A} = V^{-1} v_A, \qquad \hat{B} = V^{-1} v_B.$$

By (2): $\hat{A}_j = \Lambda_{A,j} \hat{f}_j$ and $\hat{B}_j = \Lambda_{B,j} \hat{f}_j$ for $j = 1, \ldots, d$.

Step 3: Recover the eigenvalue scalars. For each j, provided $\hat{f}_j \neq 0$ (guaranteed for generic f by Lemma 3 of [1]),

$$\Lambda_{A,j} = \frac{\hat{A}_j}{\hat{f}_j}, \qquad \Lambda_{B,j} = \frac{\hat{B}_j}{\hat{f}_j}.$$

Step 4: Reconstruct the shared secret.

$$K' \;=\; V \cdot \big(\Lambda_{A,j}\, \Lambda_{B,j}\, \hat{f}_j\big)_{j=1}^{d}. \tag{3}$$

By (2), the j-th eigencoordinate of $H_A(H_B(f))$ is $\Lambda_{A,j}\Lambda_{B,j}\hat{f}_j$, so $K' = K$ exactly.

Algorithm 1 Compute Hecke Eigenbasis (one-time public precomputation)

Input: Level N, weight k; modular forms space $M = S_k(\Gamma_0(N))$
Output: $d \times d$ change-of-basis matrix V and its inverse V^{-1}

```
cols ← []
for each newform orbit [f_ℓ] with eigenvalue field K_ℓ, degree e_ℓ do
    B ← ⌊k(N + 1)/12⌋ + 1                                  ▷ Sturm bound
    Compute q-expansion of f_ℓ over K_ℓ to precision ≥ B
    Solve A_ℓ c = y_ℓ over K_ℓ   ⇒   c ∈ K_ℓ^d
    for each embedding φ : K_ℓ ↪ L_ℓ do
        cols.append(φ(c))
    end for
end for
V ← [ cols[0] | ··· | cols[d − 1] ]
return V, V^{-1}
```

Algorithm 2 Recover Shared Secret (per-session attack)

Input: Coordinate vectors $v_f, v_A, v_B \in \mathbb{Q}^d$ of $f, H_A(f), H_B(f)$; precomputed V, V^{-1}
Output: Coordinate vector $K' \in \mathbb{Q}^d$ of the shared secret $H_A(H_B(f))$

1: $\hat{f} \leftarrow V^{-1}v_f;\ \hat{A} \leftarrow V^{-1}v_A;\ \hat{B} \leftarrow V^{-1}v_B$ ▷ project into eigenbasis
2: **for** $j = 1, \ldots, d$ **do**
3: **if** $\hat{f}_j = 0$ **then abort** ▷ f not generic; choose different f
4: **end if**
5: $\Lambda_{A,j} \leftarrow \hat{A}_j/\hat{f}_j;\ \Lambda_{B,j} \leftarrow \hat{B}_j/\hat{f}_j$
6: $p_j \leftarrow \Lambda_{A,j} \cdot \Lambda_{B,j} \cdot \hat{f}_j$
7: **end for**
8: $K' \leftarrow V \cdot (p_1, \ldots, p_d)^\top$ ▷ map back to standard basis
9: **return** K'

Theorem 3. *Given the public values $(N, k, f, H_A(f), H_B(f))$, a passive adversary can recover the shared secret $K = H_A(H_B(f))$ exactly, with the following costs:*

- ***One-time precomputation*** *(public, independent of r and s): $\widetilde{O}(B \cdot d^3)$ arithmetic operations in $\mathbb{Q}$, where $B = O(kN)$ is the Sturm bound.*
- ***Per-session attack****: $O(d^2)$ arithmetic operations in the number field K.*

The attack succeeds with probability 1 over the uniform random choice of f (in the sense of Lemma 3 of [1]).

Proof. Correctness. By (2), for any Hecke operator H acting on a vector $u = \sum_j \hat{u}_j v_j$ expressed in the eigenbasis, we have $H(u) = \sum_j \Lambda_j \hat{u}_j v_j$, i.e. H acts as component-wise multiplication by the eigenvalue vector. Applying this to $H_A(H_B(f))$ and using commutativity,

$$H_A(H_B(f)) = V \cdot \big(\Lambda_{A,j} \cdot (\Lambda_{B,j} \cdot \hat{f}_j)\big)_{j=1}^d = V \cdot \big(\Lambda_{A,j}\Lambda_{B,j}\hat{f}_j\big)_{j=1}^d.$$

This equals (3), so $K' = K$.

Precomputation cost (Algorithm 1). Let the Galois orbits have degrees $e_1, \ldots, e_r$ with $\sum_\ell e_\ell = d$. For orbit ℓ, the dominant cost is solving the $B \times d$ system $A_\ell c = y_\ell$ over K_ℓ (which has degree e_ℓ over $\mathbb{Q}$). Gaussian elimination on an $m \times n$ system costs $O(n^2 m)$ arithmetic operations in the base field; here $m = B$, $n = d$, base field $= K_\ell$, giving $O(d^2 B)$ operations in K_ℓ. Translating to $\mathbb{Q}$-operations, each K_ℓ-operation (multiplication or division) involves degree-e_ℓ polynomial arithmetic, costing $\widetilde{O}(e_\ell)$ $\mathbb{Q}$-multiplications (using fast polynomial multiplication). Hence orbit ℓ costs $\widetilde{O}(e_\ell \cdot d^2 B)$ rational operations. Summing over all orbits,

$$\sum_{\ell=1}^{r} \widetilde{O}(e_\ell \cdot d^2 B) \;=\; \widetilde{O}\left(d^2 B \sum_{\ell=1}^{r} e_\ell\right) \;=\; \widetilde{O}(d^3 B).$$

Since $B = O(kN)$ and $d = \Theta(N)$ (Proposition 1), both B and d are $\Theta(N)$, so the total is $\widetilde{O}(N^4)$ in terms of the level; we state it as $\widetilde{O}(d^3 B)$ to make explicit the separate roles of the space dimension and the Sturm bound. The Galois-conjugate step (applying e_ℓ embeddings to a vector in K_ℓ^d) costs $\widetilde{O}(e_\ell d)$ per orbit, totalling $\widetilde{O}(d^2)$: negligible. The final $d \times d$ matrix inversion V^{-1} over the compositum costs $\widetilde{O}(d^\omega)$ with $\omega < 2.373$ [5]; for the parameter sets of [1] we have $d \leq 7$ so this is a one-time constant-time computation, subdominant to the $\widetilde{O}(d^3 B)$ term.

Per-session cost (Algorithm 2). The per-session work is:

- Three matrix-vector products $V^{-1}v$ of size $d \times d$ over K: $3d^2$ multiplications and $3d(d-1)$ additions in K.
- $2d$ scalar divisions in K (Step 3).
- d scalar multiplications in K (building p_j).
- One matrix-vector product Vp of size $d \times d$ over K: d^2 multiplications in K.

Total: $4d^2 + 3d = O(d^2)$ arithmetic operations in K. The parameters r (pool size) and s (number of Hecke factors) do not appear on any line of either algorithm.

Remark 2. The parameters r and s appear nowhere in Algorithms 1–2. Increasing r from 12 to 128 (as proposed in [1]) has strictly zero effect on the adversary's running time; it only increases the honest parties' keygen cost.

5 Experimental Results

We implemented the scheme and attack in SageMath 10.7 [6] on a Linux workstation (Intel x86-64, kernel 6.8). All runs use weight $k = 2$, pool size $r = 20$, and a fixed random seed (`PYTHONHASHSEED=0`). Table 1 reports timings for the parameter sets from the paper's benchmark table.

Table 1. Attack performance against all paper parameter sets. "Precomp" is the one-time eigenbasis computation (Algorithm 1, independent of r and s); "Recovery" is the per-session attack (Algorithm 2). All runs: $K' = K$.

N	k	d	s	r	Keygen (ms)	Precomp (ms)	Recovery (ms)	$K' = K$
23	2	2	3	20	10.6	288	4	✓
37	2	2	3	20	10.2	97	2	✓
53	2	4	5	20	6.2	221	11	✓
61	2	4	7	20	8.9	167	9	✓
97	2	7	10	20	9.3	478	298,010*	✓

* The $N = 97$ recovery time is an implementation artefact, not an algorithmic one: the per-session work is dominated by exact arithmetic in Sage's generic number-field backend for the compositum of two degree-3 and one degree-5 eigenvalue fields (total degree 11). The algorithmic cost remains $O(d^2)$ K-operations; moving to a specialised number-field library (such as Pari/GP or FLINT) would be expected to reduce the wall-clock cost by orders of magnitude, although we have not benchmarked this.

Impact of r. As noted in Remark 2, r does not enter the attack. Running the experiment with $r = 128$ produces identical precomputation and recovery times; only keygen grows slightly.

6 Rebuttal of the "Underdetermination" Argument

The paper [1] considers a "natural attack" via the eigenbasis and argues that it is fundamentally limited by an underdetermined subset-product system. We identify the precise gap.

The paper's argument. Writing $f = \sum_j c_j f_j$ and $H_A(f) = \sum_j \Lambda_{A,j} c_j f_j$, the attack as framed by [1] aims to find $x \in \{0,1\}^t$ such that $\Lambda_{A,j} = \prod_{i=1}^{t} \lambda_j(M_i)^{x_i}$, where $\{M_1, \ldots, M_t\}$ is the full set of $\binom{r}{3}$ candidate 3-squarefree integers. The paper concludes this is a highly underdetermined subset-product system because $d \ll t$.

Where the argument fails. The observation is correct for recovering the *factorization* $(N_1, \ldots, N_s)$. However, the security of Hecke-KE requires *Hecke-CDH* to be hard [1]: given $(f, H_A(f), H_B(f))$, *compute* $H_A(H_B(f))$. Our attack solves Hecke-CDH directly and does not attempt to recover the factorization.

By Theorem 3, the scalars $\Lambda_{A,j}$ are uniquely recovered by d divisions in Algorithm 2. No subset-product search is needed. Moreover, the eigenbasis V is determined entirely by (N, k) and is computable by anyone using SageMath [6], PARI/GP, or Magma in a few hundred milliseconds.

The logical gap is the implicit assumption that *"the only way to compute $H_A(H_B(f))$ is to first recover the factorization $(N_1, \ldots, N_s)$."* This is false: the eigenbasis V exposes the entire Hecke-CDH computation as a sequence of d independent scalar equations, each solved by a single division. The attack exploits the algebraic structure of the Hecke algebra — specifically, the simultaneous diagonalizability of Corollary 1 — which is invisible to the underdetermination argument.

Corollary 2. *The Hecke-CDH problem is easy (polynomial time) for every parameter set proposed in [1]. Consequently, Hecke-KE does not achieve Hecke-DDH-based session-key indistinguishability, regardless of the hardness of the Hecke problem.*

Proof. Theorem 3 provides a deterministic polynomial-time algorithm computing $K = H_A(H_B(f))$ from $(f, H_A(f), H_B(f))$ whenever f is generic (Lemma 3 of [1]), which is satisfied by the protocol's distinguishing form. A distinguisher against Hecke-DDH reads the challenge $(f, H_A(f), H_B(f), z)$, recovers K' via Theorem 3, and outputs 1 iff $z = K'$. Its distinguishing advantage is 1.

7 Complexity Analysis and Unfixability

7.1 Extension to Arbitrary Level

The attack does not use primality of N anywhere. For composite level N, the Atkin–Lehner–Li theory gives a full oldform decomposition:

$$S_k(\Gamma_0(N)) \;=\; S_k(\Gamma_0(N))^{\mathrm{new}} \;\oplus\; \bigoplus_{\substack{M\mid N\\ M<N}} \bigoplus_{\ell\mid(N/M)} \alpha_\ell(S_k(\Gamma_0(M))^{\mathrm{new}})\,, \tag{4}$$

where $\alpha_\ell(f)(\tau) = f(\ell\tau)$ is the standard level-raising map.

Proposition 2. *For any $N \geq 1$ and any prime p with $\gcd(p, N) = 1$, all Hecke operators $\{T_p : p \nmid N\}$ are simultaneously diagonalizable over an explicitly computable number field K_N.*

Proof. For $p \nmid N$, the operator T_p is self-adjoint on $S_k(\Gamma_0(N))$ with respect to the Petersson inner product (Theorem 1(3)), hence diagonalizable with real eigenvalues. The family $\{T_p : p \nmid N\}$ is pairwise commuting (Theorem 1(1)), so by the spectral theorem they admit a common eigenbasis. The field K_N is the compositum of the eigenvalue fields of all newforms appearing in (4); these arise at every level $M \mid N$, so in practice one enumerates $S_k(\Gamma_0(M))^{\mathrm{new}}$ for each divisor $M \mid N$ and takes the compositum of all resulting eigenvalue fields.

Remark 3. For $p \mid N$, the Atkin–Lehner operators W_p (not the Hecke operators T_p) are self-adjoint, and T_p may not be diagonalizable on the old subspace. This is irrelevant since the protocol requires all pool primes to be coprime to N.

7.2 Galois-Orbit Decomposition

Proposition 3. *Let $S_k(\Gamma_0(N)) = \bigoplus_{\ell=1}^{r} V_\ell$ be the decomposition into Galois-stable newform orbits, with $\dim V_\ell = e_\ell$ and $\sum_\ell e_\ell = d$. The attack on each session decomposes into r independent eigenvalue recoveries, the ℓ-th operating entirely over the eigenvalue field K_ℓ. The per-session cost is $\sum_\ell O(e_\ell^2) = O(d^2)$ field operations in total.*

Corollary 3. *The attack runs in time polynomial in $d = \dim S_k(\Gamma_0(N))$ and $\log N$. In particular, no compositum-degree blow-up occurs: the per-session cost $O(d^2)$ counts operations in the individual K_ℓ, not in K.*

7.3 Unfixability

Theorem 4. *For every level $N \geq 1$ and weight $k \geq 2$, there is a passive adversary against Hecke-KE whose per-session running time is $O(d^2)$ field operations, independent of r and s, and whose public precomputation is $\widetilde{O}(B \cdot d^3)$ rational operations. Consequently, no parameter choice (N, k, r, s) makes Hecke-KE both secure against this adversary and implementable.*

Proof. Independence from r and s. By Theorem 3, the per-session attack cost is $O(d^2)$ and the one-time precomputation cost is $\widetilde{O}(B \cdot d^3)$, where $B = O(kN)$. Neither bound references r or s; increasing either parameter has zero effect on the adversary.

No secure and implementable parameter choice. For the scheme to withstand a λ-bit adversary, the per-session attack cost $O(d^2)$ must exceed 2^λ, requiring $d = \Omega(2^{\lambda/2})$.

For prime level N, Proposition 1 gives $d = g(X_0(N)) = \Theta(N)$ (since the elliptic-point counts ν_2, ν_3 are bounded by 2), so $d = \Omega(2^{\lambda/2})$ forces $N = \Omega(2^{\lambda/2})$. The honest party's public key is then a vector in $\mathbb{Q}^d$ with $d = \Omega(2^{\lambda/2})$ rational coordinates. As noted in the coefficient-domain paragraph of Sect. 1, these coordinates are in general non-integer rationals whose bit-length grows with the heights of the Hecke eigenvalues. Even charging only a single bit per coordinate, encoding the public key requires $\Omega(2^{\lambda/2})$ bits. For $\lambda = 128$ this is $\Omega(2^{64})$ bits, beyond 10^9 terabytes — far outside any implementable system.

Hence there is no (N, k) that simultaneously makes the attack infeasible and keeps the honest protocol implementable.

Remark 4 (Quantum security). The attack is a purely classical, deterministic linear-algebra computation. Hecke-KE is broken classically; the post-quantum framing of [1] does not shield the scheme.

Remark 5 (Relation to the Hecke problem). Our attack recovers the shared secret without recovering the secret factorization $(N_1, \ldots, N_s)$. The hardness of the Hecke problem is entirely irrelevant to the security of the key exchange.

8 Conclusion

Hecke-KE is broken. The Hecke algebra is finite-dimensional, commutative, and semisimple, with a completely explicit and publicly computable eigenbasis. Diagonalising in this basis reduces Hecke-CDH to d independent scalar equations, each solved by a single division.

By Theorem 4, the attack applies uniformly to every level N and weight k, running in time polynomial in $d = \dim S_k(\Gamma_0(N))$. Since every implementable choice of (N, k) yields polynomially small d, no increase in N or k can separate attacker from honest parties. Hecke-KE is therefore broken unconditionally across its entire design space.

Code availability. A complete SageMath 10.7 implementation of both the scheme and the attack is available as `hecke_ke_break.py`, with a command-line interface for arbitrary (N, k, r, s) parameter choices, and a three-part step-by-step tutorial (`part1_setup.sage`, `part2_attack.sage`, `part3_recover.sage`).

References

1. Li, T.: Modular forms and Hecke operators for post-quantum cryptography. Preprint, University of Manchester (2025)
2. Atkin, A.O.L., Lehner, J.: Hecke operators on $\Gamma_0(m)$. Math. Ann. **185**, 134–160 (1970)
3. Diamond, F., Shurman, J.: A First Course in Modular Forms. Graduate Texts in Mathematics, vol. 228. Springer, New York (2005)
4. Stein, W.A.: Modular Forms, a Computational Approach. Graduate Studies in Mathematics, vol. 79. American Mathematical Society, Providence (2007)
5. Alman, J., Williams, V.V.: A refined laser method and faster matrix multiplication. In: Proc. 32nd ACM-SIAM Symposium on Discrete Algorithms (SODA), pp. 522–539 (2021)
6. The Sage Developers: SageMath, the Sage Mathematics Software System (Version 10.7). https://www.sagemath.org (2024)